

YOUNGSEO SON

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PROFESSIONAL SUMMARY

Staff NLP / Conversational AI scientist with 10+ years of NLP research and 5+ years building production LLM systems. Currently driving AI model research direction, end-to-end implementation, and statistical validation of LLM fine-tuning pipelines (SFT, DPO) across proprietary LLMs (Gemini) and open-source SLMs (Gemma 4, Llama 3 ~4B), and RAG-based healthcare phone call automation at Infinitus Systems. Previously ML Research Scientist at Meta AI, where I designed and deployed content quality and integrity models at platform scale. Published at ACL, NAACL, EMNLP, and Psychological Medicine; DARPA SocialSim collaborator (PNNL). Expertise spans conversational AI, LLM fine-tuning (SFT/DPO/RLHF) across proprietary and open-source models, in-context and online learning with reflective prompt optimization, multimodal systems, and NLP for high-stakes domains (healthcare, AI safety, mental health).

SKILLS

LLM & NLP: PyTorch · HuggingFace · SFT · DPO · RLHF · In-context learning · Online learning · Reflective prompt optimization · RAG · DSPy · LLM-as-judge · GPT-4o / Gemini / Gemma / Llama APIs · BERT / transformer fine-tuning · TensorFlow · scikit-learn

Data & Infra: BigQuery · Spark · Presto · MySQL · MongoDB · GCP (Vertex AI, VMs) · Kubernetes · Ubuntu server management

Languages: Python (primary) · Java

WORK EXPERIENCE

Infinitus Systems Mar 2024 – Present

Staff Applied NLP Scientist (Member of Technical Staff)

- Drove research direction and implemented end-to-end LLM improvement pipelines: SFT and DPO fine-tuning on proprietary LLMs (Gemini) and open-source SLMs (Gemma 4, Llama 3 ~4B), augmented with in-context + online learning and reflective prompt optimization; led model validation and statistical analysis across **[X] healthcare clients / Y call types**.
- Built real-time RAG conversational AI system for end-to-end healthcare phone call automation; achieved **[X% task-completion accuracy / Y% reduction in average handle time]**. (ACL 2026 Industry)
- Developed LLM-as-judge framework for call failure analysis with knowledge injection, enabling AI review of **[X]K+ calls/day** and improving error detection accuracy by **[Y%]**. (ACL 2025)
- Built hybrid multimodal pipeline combining GPT/Gemini LLMs with fine-tuned Whisper audio models for voice activity detection, user frustration recognition, and call failure classification. (NAACL 2024)
- Iteratively evaluated and integrated state-of-the-art LLMs and SLMs to optimize healthcare automation efficiency; contributed to **[X%]** improvement in [key metric].

Mar 2023 – Mar 2024

Senior Applied NLP Scientist (Member of Technical Staff)

- Joined as senior scientist; promoted to Staff within one year based on research impact, system ownership, and leadership of model development and validation efforts.

Meta Feb 2021 – Jan 2023

ML Research Scientist

- Designed comment quality metrics and fine-tuned NLP ranking models to surface high-quality conversations across Facebook public posts, Reels, and live video, serving **[hundreds of millions of]** users.
- Fine-tuned and deployed content integrity models at platform scale to detect hate speech, harassment, and bullying; achieved **[X%] reduction in violating content** across **[Y]B+ content items reviewed**.
- Ran A/B experiments to optimize comment ranking models across multiple product surfaces; model improvements shipped to **[X]M+ daily active users**.

Facebook Summer 2020

ML PhD Software Engineer Intern | Search Integrity

- Built graph-based NLP model to improve search result quality; conducted data mining to surface ranking signals and reduce policy-violating search results.

Pacific Northwest National Laboratory (PNNL) Summer 2019

Data Scientist PhD Intern | DARPA SocialSim

- Developed SocialSim modules to analyze cross-platform information / graph evolution and detect coordinated inauthentic behavior and misinformation spread on Twitter, Reddit, and GitHub.

Stony Brook University May 2017 – Dec 2020

Research Assistant · NLP / Computational Social Science

- Developed NLP models correlating language with mental/physical health outcomes for WTC 9/11 responders (LDA topic clustering, discourse parsing, sentiment analysis); published in *Psychological Medicine*.
- Ranked #1 on NAACL 2019 CLPsych Shared Task (Task B): suicide risk assessment from Reddit using user-factor-adapted BERT with post-level attention.

World Well-Being Project, University of Pennsylvania Summer 2016

Research Scientist Intern

- Led development of a discourse relation parser capturing counterfactual thinking in social media text; published at ACL 2017.

EDUCATION

Stony Brook University Aug 2015 – Dec 2020

Doctor of Philosophy in Computer Science · GPA 3.89 / 4.00

Stony Brook University Aug 2013 – May 2015

B.S. Computer Science, Summa Cum Laude, Honors Program · GPA 3.93 / 4.00

Ajou University, South Korea Mar 2009 – Aug 2013

B.E. Computer Engineering · GPA 4.36 / 4.50 (Major 4.43 / 4.50)

SELECTED PUBLICATIONS

Full list: scholar.google.com/citations?user=ILZe83UAAAAJ

Transfer Learning for Generalizable Automated LLM Improvement Pipeline for IVR Navigation. *Youngseo Son et al.* **ACL 2026 Industry Track**, 2026.

Auto Review: Second Stage Error Detection for Highly Accurate Information Extraction from Phone Conversations. *Ayesha Qamar, Arushi Raghuvanshi, Conal Sathi, Youngseo Son.* **ACL 2025 Industry Track**, 2025.

Multimodal Contextual Dialogue Breakdown Detection for Conversational AI Models. *Md Messal Monem Miah, Ulie Schnaithmann, Arushi Raghuvanshi, Youngseo Son.* **NAACL 2024 Industry Track**

Discourse Relation Embeddings: Representing the Relations between Discourse Segments in Social Media. *Youngseo Son, Vasudha Varadarajan, H. Andrew Schwartz.* **EMNLP 2022 (UM-IoS Workshop)**

Predicting Adolescent Depression and Anxiety from Multi-wave Longitudinal Data using Machine Learning. *Mariah T. Hawes, H. Andrew Schwartz, Youngseo Son, Daniel N. Klein.* **Psychological Medicine**, 2022.

World Trade Center Responders in Their Own Words: Predicting PTSD Symptom Trajectories with AI-Based Language Analyses. *Youngseo Son, Sean A. P. Clouston, Roman Kotov, Johannes C. Eichstaedt et al.* **Psychological Medicine**, 2021.

Suicide Risk Assessment with Multi-level Dual-Context Language and BERT. *Matthew Matero, Akash Idnani, Youngseo Son et al.* **NAACL 2019, CLPsych Workshop**

SERVICE & RECOGNITION

Area Chair, Industry Track — EACL 2027 2027

Workshop Organizer — TRUSTMORE 2026 (1st Intl. Workshop on Trustworthy Multimodal Agents) — IEEE BigData 2026 2026

Outstanding Reviewer — ACL 2021 Jul 2021

Program Committee / Reviewer — ACL, EMNLP Ongoing

Special CS Department Chair Fellowship — Stony Brook University 2015

Computer Science Award of Honor — Stony Brook University 2015